Petitioner/Beneficiary: Sandra Rebecca Babyale

Petition: I-140, EB-2 National Interest Waiver- INA ξ 203 (b) (2) (B)

(Original Submission)

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RE: I-140 Immigrant Petition for Alien Worker - A member of the professions holding an advanced degree or an alien of exceptional ability - National Interest Waiver

Petitioner/Beneficiary: Sandra Rebecca Babyale, M.Sc.
Nature of submission: Original Submission
Type of Petition: I-140, EB-2 National Interest Waiver
Classification Sought: INA §203(b)(2)(B) National Interest Waiver

Dear USCIS Officer:

I, Sandra Rebecca Babyale, respectfully submit this petition for classification as a member of the professions holding an advanced degree and request a National Interest Waiver of the job offer and labor certification requirement pursuant to section 203(b)(2)(B)(i) of the Immigration and Nationality Act and the precedent decision in *Matter of Dhanasar, 26 I&N Dec. 884 (AAO 2016).*

I hold a Master of Science in Mathematical Sciences with Distinction from the African Institute for Mathematical Sciences (AIMS) Rwanda, and I am a Ph.D. candidate in Computing with a specialization in Computational Mathematics, Science, and Engineering at Boise State University, maintaining a summa cum laude GPA of 3.968/4.0. Evidence of my advanced degrees is included in the supporting exhibits.

My proposed endeavor is to advance wildfire smoke transport forecasting in the United States through improved data assimilation methods, model error covariance estimation techniques, scientific machine learning approaches, and high-performance atmospheric transport modeling. Wildfire smoke has become one of the most rapidly intensifying environmental hazards in the United States, and enhancing national forecasting systems is essential for protecting public health, supporting emergency response, and improving air quality management across the nation.

This petition demonstrates that my proposed endeavor has substantial merit and national importance, that I am well-positioned to advance this work, and that, on balance, it is in the national interest of the United States to waive the job offer and labor certification requirement. This petition letter is submitted in support of my Form I-140. Specifically, the submitted evidence will prove the following:

1. **I am a member of the professions holding an advanced degree:** I hold a Master of Science in Mathematical Sciences with Distinction and am currently a Ph.D. candidate in Computing with an emphasis in Computational Mathematics, Science, and Engineering. (Please refer to Chapter 1).

2. **My proposed endeavor has both substantial merit and national importance:** My proposed endeavor is to develop advanced data assimilation methods, model error estimation techniques, and scientific machine learning frameworks to improve wildfire smoke transport forecasting accuracy. This research directly addresses a major national environmental and public health crisis affecting millions of Americans annually. (Please refer to Chapter 2).

3. **I am well-positioned to advance the proposed endeavor:** My education, skills, knowledge, record of success, NSF-funded research contributions, peer-reviewed publication, and demonstrated interest from federal agencies and independent experts position me to make significant contributions in this field. (Please refer to Chapter 3).

4. **On balance, it would benefit the United States to waive the requirements of a job offer and, thus, of a labor certification.** (Please refer to Chapter 4).

Your assistance in this case will be appreciated.

Chapter One

Advanced Degree Professional

To satisfy the underlying requirements for this classification, I, Sandra Rebecca Babyale, am submitting evidence that I am a member of the professions holding an advanced degree. As defined in the implementing regulations, an advanced degree "[m]eans any United States academic or professional degree or a foreign equivalent degree above that of baccalaureate" (8 CFR §204.5(k)(2)) and must be demonstrated by "[a]n official academic record showing that the alien has a United States advanced degree or a foreign equivalent degree" (8 CFR §204.5(k)(3)(i)(A)).

I am an expert in computational mathematics with advanced training in data assimilation, geophysical inverse problems and scientific machine learning, holding two degrees. I obtained a Bachelor of Science with Education in Physics and Mathematics with a CGPA of 4.20/5.0 from Busitema University, Uganda *(See Exhibits B1, B2, B3)* and a Master of Science in Mathematical Sciences with Distinction from the African Institute for Mathematical Sciences (AIMS) Rwanda. As evidence of my M.Sc., I am submitting copies of my admission letter, diploma, transcripts and thesis *(See Exhibits B4, B5, B6, D22)*. Since I completed my master's degree outside the United States of America, I am also submitting a detailed advisory evaluation of my educational credentials *(See Exhibit B7)*. Therefore, I am qualified as a member of the professions holding an advanced degree.

Professor Jodi Mead confirms the exceptional nature of AIMS training: ***"I first met Sandra at the African Institute for Mathematical Sciences (AIMS), a network of highly selective graduate centers across Africa known for their rigorous mathematical and computational training. Recognizing her exceptional ability, I recruited her to Boise State to pursue a Ph.D. in Computing." (See Exhibit A2)***

I am currently a Ph.D. candidate in Computing with a specialization in Computational Mathematics, Science, and Engineering at Boise State University, where I am on track to graduate summa cum laude with a GPA of 3.968/4.0 *(See Exhibits B8, B9, B10)*. My dissertation research focuses on developing advanced data assimilation methods and model error estimation techniques to improve wildfire smoke transport forecasting *(See Exhibit D21)*. This work has profound implications for public health and environmental protection in the United States, as wildfire smoke has become one of the most rapidly intensifying environmental hazards affecting millions of Americans annually.

Chapter Two
Substantial Merit and National Importance

The first prong of the NIW framework, as established in Dhanasar, requires that I demonstrate that my proposed endeavor has substantial merit and national importance. As an initial matter, this prong "focuses on the specific endeavor that the foreign national proposes to undertake," so the proposed endeavor must be clearly defined so that its merit and importance can be evaluated.

As an expert in Computational Science, **my proposed endeavor is to improve the accuracy and reliability of wildfire smoke transport forecasts in the United States by advancing the scientific foundations of model error estimation, variational data assimilation techniques, and state-of-the-art machine learning frameworks for large-scale atmospheric transport models.** Wildfire smoke forecasting is a field that directly influences public health protection, air quality decision making, emergency response, transportation management, and economic stability. Federal agencies have documented significant limitations in current prediction systems and have emphasized the need for improved modeling and advanced computational techniques.

Machine learning and advanced modeling and simulation, which are major components of my proposed research, are federally designated Critical and Emerging Technologies. The National Science and Technology Council's February 2024 Critical and Emerging Technologies List identifies artificial intelligence which includes machine learning and deep learning, as one of 18 technology areas of particular importance to U.S. national security *(See Exhibit C6, pp. 2, 4)*. The same document identifies advanced computing, encompassing advanced modeling and simulation capabilities, as another critical technology area *(See Exhibit C6, p. 2)*.

Because my proposed endeavor integrates both artificial intelligence and advanced modeling and simulation into wildfire smoke forecasting, it operates at the intersection of two federally designated critical technologies. This gives the endeavor significant scientific and national importance, as recognized at the highest levels of federal policy. The following sections demonstrate that the proposed endeavor has substantial intrinsic scientific merit and clear national importance.

2.1 My Proposed Endeavor

2.1.1 The National Challenge

The United States faces a growing national challenge from wildfire smoke, which affects public health, transportation, air quality management, infrastructure resilience, and economic stability. A comprehensive 2023 analysis by the U.S. Congress Joint Economic Committee found that climate-exacerbated wildfires cost the nation massive economic losses annually. Specifically, the Joint Economic Committee reports that wildfires in the United States cause ***"between $394 billion and $893 billion dollars in damages annually, which is equivalent to between 2-4% of U.S. GDP" (See Exhibit C1, JEC October 2023, p. 2).*** This Congressional analysis further finds that ***"costs from short and long-term exposure to wildfire smoke are estimated at $117.5 - 202.5 billion" (See Exhibit C1, p. 3)***, demonstrating that smoke exposure alone accounts for a substantial portion of these massive national costs.

The Congressional committee concluded that ***"the immense cost of wildfires—both the human toll and the economic damages—requires government action" (See Exhibit C1, p. 2)***, explicitly calling for federal response to what the committee characterizes as a ***"growing threat to the health and well-being of communities across the country" (See Exhibit C1, p. 1).***

Federal agencies including the United States Government Accountability Office (GAO), the National Oceanic and Atmospheric Administration (NOAA), the Environmental Protection Agency (EPA), the Department of the Interior (DOI), the Centers for Disease Control and Prevention (CDC), and the Wildland Fire Mitigation and Management Commission have documented that wildfire smoke events are increasing in scale and intensity and that the systems used to predict smoke dispersion require major scientific and technical improvements.

Central to this challenge is the need for accurate, reliable, and timely wildfire smoke forecasts. These forecasts depend on atmospheric transport models that integrate large volumes of remote sensing and in situ observations. However, current forecasting systems face substantial limitations.

2.1.2 Current Forecasting Limitations

The latest peer-reviewed research confirms that ***"improvement of plume dynamics modeling capabilities is urgently required, and yet, it is a great challenge to improve the smoke modeling needed to support operational decision-making for assessing and mitigating smoke impacts, especially in regions with complex terrain and urban areas" (See Exhibit C16, Liu et al. 2025, p. 3)***. Federal multi-agency studies have documented the specific barriers preventing accurate smoke predictions. A comprehensive NOAA, NASA, and Naval Research Laboratory multi-agency ensemble study explains that ***"due to large uncertainties in fire emission, transport, and chemical transformation, it remains challenging to accurately predict air quality during wildfire events" (See Exhibit C8, BAMS Multi-Agency Study, Abstract).***

Empirical model evaluations demonstrate the severity of these limitations. A NASA-led study evaluating twelve atmospheric models found that ***"biomass burning emissions and smoke remain among the largest sources of uncertainties in air quality forecasts" (See Exhibit C9, Ye et al. 2021, Abstract).*** The study revealed that ***"the comparison of smoke organic carbon (OC) emissions suggests a large range of daily totals among the models with a factor of 20 to 50" (See Exhibit C9, Abstract),*** demonstrating extreme variability even among state-of-the-art operational systems. Critically, ***"the evaluation of smoke AOD (sAOD) forecasts suggests overall underpredictions in both the magnitude and smoke plume area for nearly all models" (See Exhibit C9, Abstract),*** representing systematic forecast failures across all operational forecasting systems. Furthermore, the study found that ***"discrepancies in model performance for surface $PM_{2.5}$ and AOD ... cannot be compensated by solely adjusting the smoke emissions but are more attributable to model representations of plume injections" (See Exhibit C9, Abstract),*** highlighting the fundamental limitations requiring advanced methods beyond simple emission scaling.

Independent peer-reviewed forecast evaluations confirm these problems persist. A study published in Environmental Health Perspectives found only ***"modest agreement between forecasts and observations"*** *with a correlation of just 0.4* ***(See Exhibit C24, Yao et al. 2013, p. 1142, 1145).*** The authors explain that ***"disagreement between BlueSky forecasts and observed data could come from the limitations of the BlueSky system, including its inability to predict***

smoke from fires outside of the modeling domain, and uncertainty in the input meteorology and/or fire information" (See Exhibit C24, p. 1145).

These forecasting limitations have persisted for more than two decades. A 2001 federal multi-agency review documented that ***"the greatest uncertainty in forecasting air quality impacts from natural disasters such as volcanic eruptions and wildfires is associated with the source term"*** and that ***"a better quantification of emission rates could dramatically improve the reliability and therefore the usefulness of such forecasts" (See Exhibit C12, NOAA/CENR 2001, p. 16).*** More than twenty years later, a comprehensive 2025 review confirmed that ***"model fire emissions are a major uncertainty" (See Exhibit C16, Liu et al. 2025, p. 14),*** representing persistent fundamental gaps in operational forecasting systems.

2.1.3 The Role of Data Assimilation

Federal atmospheric science research confirms that numerical aerosol prediction ***"is at a crossroads"*** and ***"it has experienced quick progress in recent years due to the availability of aerosol models, aerosol satellite observations, data assimilation techniques, and the know-how of numerical weather prediction" (See Exhibit C15, Benedetti et al., p. 10635).*** However, translating this scientific progress into operational forecasting capabilities requires addressing persistent mathematical challenges in model error estimation.

The United States Government Accountability Office describes the critical importance of advanced data assimilation methods:

"Improving data assimilation. Machine learning can speed up data assimilation—the process of updating forecast models by repeatedly integrating the most current environmental observations to best represent the initial conditions for a prediction … Faster data assimilation could allow models to use more data, such as satellite data that is significantly underutilized. This in turn would provide a more accurate picture of current conditions and should result in a more accurate forecast." (See Exhibit C3, GAO Report 2023, p. 20)

This federal finding demonstrates that operational forecasting systems do not currently make full use of the available observational record, with satellite data remaining ***"significantly underutilized"*** despite its potential to improve forecast accuracy.

The GAO report identifies machine learning as critical for accelerating data assimilation to enable fuller use of satellite observations. Current operational systems face severe computational time constraints during active wildfire events. Federal emergency response protocols require smoke forecasts within 6-12 hour windows to support evacuation planning, air quality alerts, and resource allocation decisions.

My physics-informed machine learning research directly addresses this operational bottleneck. While classical iterative methods require hours of computation for parameter estimation, trained DeepONet architectures reduce this to seconds, enabling the rapid forecast updates essential for emergency response. This capability transforms the GAO's finding about "significantly underutilized" satellite data from a theoretical possibility into an operational reality by making real-time assimilation computationally feasible.

Federal researchers at the Naval Research Laboratory confirm that ***"data assimilation is an essential component in skillful atmospheric models of all kinds, but it demands accurate numerical estimates of the uncertainty in both the forecast model and the relevant observations" (See Exhibit C10, Hyer et al., p. 2).*** However, current operational forecasting systems lack these accurate uncertainty estimates, representing a critical gap between available satellite and in situ observations and effective operational implementation. One major limiting factor is the absence of accurate and computationally stable model error covariance representations, which are required for advanced assimilation techniques.

2.1.4 The Model Error Challenge

Multiple federal studies identify model uncertainty as a critical challenge *(See Exhibits C8, C10, C9, C16, C14).* Federal model inter-comparison studies reveal the magnitude of this problem: for the same wildfire events in August 2019, emission estimates from different operational forecasting systems showed ***"max-to-min ratios of model emissions ... range from 20 on 7 August and 60 or more on 3 and 10 August" (See Exhibit C16, Liu et al. 2025, p. 14).*** This means that model predictions for the same fire, using the same input data, differed by factors of

20 to 60. This massive spread demonstrates the fundamental challenge: operational forecasting systems lack robust methods to quantify their own uncertainties.

The study identifies ***"the satellite products that were used to estimate the model fire emissions"*** as ***"a major uncertainty" (See Exhibit C16, p.14)***, but the 20-60x variation between models reveals that uncertainty propagation and error quantification remain poorly understood. A 2025 comprehensive review confirms that ***"the major remaining gaps are the lack of comprehensive simultaneous measurements of smoke, fuels, fire, and atmospheric interactions during wildfires, high-resolution coupled modeling systems of these components, and real-time smoke prediction capacity" (See Exhibit C16, Abstract).*** Without comprehensive simultaneous measurements of smoke plumes, models cannot properly quantify their prediction errors and properly quantifying model error remains impossible.

The systematic nature of these errors compounds forecasting challenges. Models ***"tend to overpredict plume rise for the shallower injections and underpredict for the days with deeper injections" (See Exhibit C16, p.14),*** creating predictable bias patterns that vary with atmospheric conditions. Beyond these vertical distribution errors, models struggle with spatial and temporal accuracy, ***"the biases for CO and BC simulations were higher because the model was not able to reproduce the timing, shape, and location of individual plumes over complex terrain (Rocky Mountains)" (See Exhibit C16, p.14).*** These systematic failures in capturing both injection height and plume transport patterns indicate that model error covariance structures remain poorly characterized, limiting the effectiveness of current forecasting approaches.

These uncertainties persist in operational forecasting systems. Naval Research Laboratory scientists emphasize that ***"Data assimilation is an essential component in skillful atmospheric models of all kinds, but it demands accurate numerical estimates of the uncertainty in both the forecast model and the relevant observations" (See Exhibit C10, Hyer et al.).*** Recent NOAA operational model evaluations confirm these challenges: analysis of the National Air Quality Forecast Capability found ***"substantial uncertainties in wildfire emissions and plume rise, smoke transport, and smoke plume chemistry" (See Exhibit C11, Li et al. 2025, p. 1646)*** with the system showing that it ***"does not improve model performance for $PM_{2.5}$ in the vicinity and***

downwind of fire emission sources...thus indicating a focal point of model uncertainty and needed improvement" (See Exhibit C11, abstract).

2.1.5 My Research Approach

My proposed endeavor directly addresses these federally documented needs by focusing on the following areas:

1. **Improving variational data assimilation for wildfire smoke transport:** I develop new formulations and numerical strategies that incorporate realistic model error structures, improve stability, and enhance the accuracy of state estimation in highly dynamic smoke environments.
2. **Developing improved model error covariance estimation techniques:** I design non-isotropic, observation-informed, and regularization-based covariance structures for atmospheric transport models to improve forecast accuracy, quality and uncertainty quantification.
3. **Advancing scientific machine learning for smoke forecasting:** I create machine learning-assisted inverse modeling tools for estimating wildfire smoke emissions, parameters, and boundary conditions while maintaining consistency with physical laws.
    - Physics-informed architectures embedding conservation laws and atmospheric dynamics into neural network training, ensuring predictions respect fundamental transport physics
    - Operator learning methods (DeepONets) enabling rapid parameter inference from sparse, noisy satellite observations—transforming hours of classical computation into seconds of neural network inference
    - Hybrid frameworks combining my published automated covariance estimation methods with ML computational efficiency, maintaining statistical rigor while meeting operational time constraints

- Uncertainty-aware predictions providing probabilistic forecasts with quantified confidence intervals, addressing federal requirements for forecast reliability metrics in emergency management contexts

4. **Integrating satellite and ground smoke observations into transport models:** My work combines aerosol optical depth measurements, $PM_{2.5}$ station data, and remote sensing products with atmospheric transport simulations to improve prediction accuracy, addressing the GAO's finding that satellite data remains *"significantly underutilized."*
5. **Developing scalable, high-performance computing approaches:** I implement and optimize methods that support large-scale transport modeling with adaptive mesh refinement and multilevel domain structures, addressing computational challenges in atmospheric modeling where ***"some of these calculations can take a long time to run and consume large amounts of energy"(See Exhibit C3, GAO, p. 17).***

2.2 My Proposed Endeavor is of Substantial Intrinsic Merit

2.2.1 Scientific and Technical Importance of Wildfire Smoke Transport Modeling

Wildfire smoke transport modeling is a scientifically complex discipline within atmospheric science and environmental prediction. The movement of smoke depends on a combination of physical, chemical, and meteorological processes that are difficult to observe, represent, and predict. Because these processes evolve rapidly and interact across multiple spatial and temporal scales, even small errors in their representation can propagate through the forecast system, producing significant deviations in predicted smoke concentrations.

Recent evaluations of NOAA's operational National Air Quality Forecast Capability reveal the magnitude of current challenges. Research on the AQMv7 model identifies ***"substantial uncertainties in wildfire emissions and plume rise, smoke transport, and smoke plume chemistry" (See Exhibit C11, p. 1646)*** and finds that model performance remains poor for $PM_{2.5}$ concentrations in regions affected by wildfire smoke, indicating a critical focal point of model uncertainty requiring improved quantification methods. These findings demonstrate that even operational federal forecasting systems struggle to properly quantify and reduce model uncertainties, particularly in regions most affected by wildfire smoke.

The Congressional wildfire commission emphasizes the urgency and national scope of this challenge, noting that ***“the wildfire crisis in the United States is urgent, severe, and far reaching”*** and documenting that ***“wildfires are not just burning with greater severity, but are becoming an emergent threat in areas that have little or no history of wildfire.” (See Exhibit C2, Wildland Fire Mitigation Commission report WFMMC, p.1, 5).*** The commission identifies critical needs spanning ***“modernizing tools for informed decision-making”*** including improved forecasting capabilities to support public health protection and emergency response *(See Exhibit C2, p. 25)*.

NOAA's 2020 Strategic Directions Workshop identified systematic gaps across operational forecasting systems. The workshop documented that current systems face ***“unmet air quality forecasters’ requests for high resolution AQ predictions (3km or finer) and extended AQ predictions out to 5 days”*** and emphasized the need to ***“improve the diurnal cycle and plume rise for fire emissions”*** and ***“improve initialization and assimilation of AOD and NO2 data” (See Exhibit C7, NOAA Air Quality 2020 Workshop Report, p.7).*** Federal researchers from multiple agencies reached consensus that addressing wildfire smoke forecasting requires ***“comprehensive and accurate coupled earth system model for improved operational prediction”*** with properly represented model uncertainties *(See Exhibit C7, p. 7)*.

Federal research assessments document that these challenges persist across multiple modeling systems and span emission inventories, plume rise parameterizations, chemical transformation processes, and aerosol transport *(See Exhibits C8, C10, C9, C14, C16, C15)*. The complexity arises from fundamental physical processes: emissions depend on heterogeneous fuel properties that vary sharply across landscapes; plume rise depends on thermodynamic conditions that can shift rapidly; and aerosol chemistry involves complex multiphase transformations that depend on environmental conditions. Liu et al. (2025) confirm that ***“improving plume dynamics science will require the use of physics-based, coupled fire–atmosphere or fire–atmosphere–chemistry models” (See Exhibit C16, p. 16),*** establishing that current operational approaches do not adequately represent the integrated physical processes governing smoke behavior.

A major source of scientific complexity arises from the need to represent the vertical distribution of smoke. Plume injection height has a dominant influence on downstream concentrations, yet it depends on fire intensity, atmospheric stability, and local thermodynamic structure. Errors in representing vertical distribution can cause large divergences in predicted surface concentrations and regional transport patterns. Federal research confirms that current operational systems rely on simplified plume rise parameterizations that introduce significant uncertainties into smoke forecasts *(See Exhibits C10, C16)*.

Wildfire smoke modeling also requires integrating diverse observational data, including satellite aerosol optical depth, geostationary fire radiative power, lidar backscatter, and ground-level particulate matter measurements. Each observation type provides only a partial view of the plume structure, and combining them into a coherent state estimate requires advanced scientific methods. However, federal assessments have documented persistent limitations in operational systems' ability to effectively assimilate these diverse data sources, with particularly critical gaps in utilizing satellite observations for smoke forecasting *(See Exhibits C3, C10, C14)*.

The scientific literature confirms that effectively using these observations depends on proper data assimilation frameworks with accurate model error covariance representations. Advanced data assimilation techniques, particularly weak constraint four dimensional variational methods, require mathematically rigorous estimates of model error structures to properly weight observations against model predictions and correct for systematic biases. However, operational forecasting systems currently lack the computational tools and mathematical frameworks needed to implement these advanced techniques at scale *(See Exhibits C10, C11, C16)*.

Taken together, these factors confirm that wildfire smoke transport modeling is a scientifically important and technically demanding field. It involves complex physical and chemical processes, high-dimensional state spaces, intricate interactions between emissions, atmospheric conditions, and aerosol transformations, and systematic error patterns that current operational systems cannot adequately characterize or reduce. The substantial scientific merit of the proposed endeavor follows directly from this inherent complexity and the federally documented need for continued advances in the mathematical, physical, and computational foundations of smoke modeling, particularly in model error covariance estimation and advanced data assimilation methods.

2.2.2 Summary of Substantial Merit

The proposed endeavor carries substantial intrinsic scientific merit because it advances the mathematical, physical, and computational foundations of wildfire smoke transport modeling. Federal agencies have identified this field as scientifically demanding and in need of significant improvement. Multiple federal research assessments document critical gaps in coupled modeling systems, model error quantification, effective use of satellite observations, and operational data assimilation capabilities *(See Exhibits C3, C2, C5, C8, C7, C10, C9, C11, C14, C15, C16).*

By strengthening the scientific basis of smoke transport models, addressing federally documented technical gaps in model error estimation and data assimilation, and introducing innovative hybrid methods combining variational techniques with machine learning, the endeavor satisfies the requirement of substantial intrinsic merit under the first prong of the Dhanasar framework.

2.3 My Proposed Endeavor Has National Importance

The national importance of accurately estimating and forecasting wildfire smoke is profound. Wildfire smoke affects public health, economic productivity, transportation, emergency management, and essential infrastructure systems across the United States. Federal agencies have documented that smoke exposure is widespread, severe, and increasing. These institutions consistently emphasize the urgent need for more reliable smoke forecasts.

2.3.1 Public Health Importance

Wildfire smoke now represents one of the most significant and widespread public health hazards in the United States. The Congressional Joint Economic Committee found that ***"smoke from wildfires (particulate matter, nitrogen oxides, and ozone) can lead to premature mortality; increased asthma incidences, including among children; and many other negative health outcomes" (See Exhibit C1, JEC report 2023, p. 3).*** These health effects are not limited to fire-adjacent areas but affect communities nationwide due to long-range smoke transport. The National Institute of Standards and Technology (NIST) documents that ***"the health effects of a wildfire are not confined temporally or geographically to the fire event itself. Smoke from a***

fire can be carried hundreds of miles away, and the effects of smoke exposure may persist well after the fire has been extinguished. These impacts are not insignificant, nor are they simple to quantify." (See Exhibit C13, p. 23).

The Wildland Fire Mitigation and Management Commission emphasized the severity and national scope of health impacts, documenting that ***"inhalation of smoke or its byproducts has been linked to a long list of conditions including respiratory and cardiovascular diseases, cancer, and mental health issues"*** and noting that ***"recent research found that long-term smoke exposure is responsible for upwards of 6,000 additional deaths per year across the contiguous United States, the majority of which occur in the eastern half of the country" (See Exhibit C2, WFMMC, p.7).*** The Commission further emphasized that ***"firefighters in particular face prolonged exposure, the impacts of which are still poorly understood, as do agricultural workers and others who work outdoors" (See Exhibit C2, p.7).***

The Department of Interior's Wildland Fire Economic Review provides concrete documentation of smoke-related health costs. The review found that ***"the average annual cost of wild-fire smoke exposure in California was $192 million (2020 dollars, $217 million in 2022 dollars)"*** with total costs over a seven-year study period reaching ***"$1.3 billion for California alone" (See Exhibit C***5***, DOI Wildland Fire Economic Report 2023, p. 4).*** These documented costs underscore that smoke health impacts represent a quantifiable, substantial economic burden beyond the immediate human suffering.

Because smoke crosses state and regional boundaries, the resulting health effects have a national footprint affecting millions of Americans annually. Accurate smoke forecasts are essential for guiding public health measures such as issuing air quality alerts, advising schools and childcare facilities, adjusting outdoor work schedules for labor safety, informing communities about protective actions, and supporting emergency response operations.

2.3.2 Economic Impact of Wildfire Smoke

The U.S. Congress Joint Economic Committee found that wildfires ***impose between $394 billion and $893 billion in annual damages, equivalent to 2-4% of U.S. GDP (See Exhibit C1, JEC October 2023, pp. 1-2). Short and long-term exposure to wildfire smoke alone accounts for***

$117.5-202.5 billion of this burden (See Exhibit C1, p. 3). The Congressional committee further notes that ***"the total cost estimates in this report should be viewed as a likely undercount of the true total cost, as there are several costs connected to wildfires that have not yet been fully quantified by researchers" (See Exhibit C1, p. 2)***, suggesting the real economic impact may be even larger.

The Wildland Fire Mitigation and Management Commission (WFMMC) emphasizes the national scale and urgency of economic impacts: ***"The total costs of wildfire are, at present, impossible to fully tabulate, federal suppression efforts now total well over $2.5 billion per year"*** and ***'federal agencies estimate that the total cost of wildfire nation-wide is "on the order of tens to hundreds of billions of dollars per year"' (See Exhibit C2, WFMMC 2023, p. 1).*** The Commission documents that these costs extend beyond direct fire damage: ***"Much of this financial toll is borne at the community level and often quickly overwhelms local resources" (See Exhibit C2, p.1).***

These economic losses extend across multiple sectors: health care systems experience increased demand during smoke events; transportation networks face disruptions from visibility degradation; businesses suffer from closures and reduced productivity; and agriculture sustains damage from smoke exposure *(See Exhibit 23)*. The Department of Agriculture's development of the Fire Insurance Protection-Smoke Index endorsement for grape crops demonstrates federal recognition of smoke as a significant agricultural threat, with the program designed to cover ***"a portion of the deductible of the Grape Crop Provisions when the insured county experiences a minimum number of Smoke Events as determined by Federal Crop Insurance Corporation(FCIC)" (See Exhibit C21, USDA Risk Management Agency 2025).*** California legislative hearings documented that wildfire smoke during peak harvest season created ***"enormous strain"*** on agricultural operations, with farmers facing insurance premium increases from $8,000 to $36,000 and lost wages reaching as much as $50 million for farmworkers in a single county *(See Exhibit C22, California Assembly Committee on Agriculture 2020 pp. 4, 6).* The economic importance of improved forecasting is clear: better predictions enable protective actions that reduce these massive economic costs across health care, transportation, business operations, and agricultural production.

2.3.3 Federal Recognition of Research Needs

Federal agencies across multiple departments have explicitly identified the need for improved wildfire smoke forecasting. The U.S. Congress Joint Economic Committee emphasized that ***"these significant costs from wildfires motivate continued policy action to reduce the incidence of catastrophic wildfires and address their significant effects on people and the planet" (See Exhibit C1, p. 2).***

The Wildland Fire Mitigation and Management Commission issued an urgent call for modernized forecasting capabilities as part of comprehensive wildfire management. The Commission identified ***"modernizing tools for informed decision-making"*** as a critical theme, emphasizing the need for ***"science, data, and technology"*** improvements and noting that ***"decision-making must be driven by science and data that reflect the full complexity and the interdisciplinary nature of wildfire in the 21st century" (See Exhibit C2, p. 25-26).*** The Commission specifically emphasized that ***"we must embrace modern tools, including remote sensing, real-time decision support technologies, and updated modeling" (See Exhibit C2, p. 26).***

The Government Accountability Office specifically identified data assimilation improvements and machine learning integration as priorities for enhancing forecast accuracy, documenting that satellite data for hazard modeling remains ***"significantly underutilized"*** and that ***"some of these calculations can take a long time to run and consume large amounts of energy,"*** highlighting the need for computational efficiency improvements *(See Exhibit C3, GAO 2024, p. 17, 20).*

NOAA's multi-agency strategic workshop documented systematic gaps requiring immediate attention, identifying the need for ***"improving the accuracy of atmospheric composition and AQ predictions focusing on emissions modeling (wildfires and dust, and timely updates of anthropogenic sources), meteorology (e.g. PBL height, cloudiness, vertical transport), and initialization of atmospheric composition concentrations (model state) and emissions using coupled data assimilation" (See Exhibit C7, p. 7).***

2.3.4 Summary of National Importance

The national importance of my proposed endeavor is established through:

1. **Massive Economic Burden:** $394-893 billion annually (2-4% of U.S. GDP), with smoke exposure alone costing $117.5-202.5 billion, as documented by the U.S. Congress *(See Exhibit C**1**)*.

2. **Widespread Public Health Impacts:** Smoke causes premature mortality, increased asthma, cardiovascular complications, and emergency hospitalizations affecting millions of Americans, with documented costs of $1.3 billion in California alone over seven years, as confirmed by federal commission and departmental reviews *(See Exhibits C**2**, C5, C**1**)*.

3. **Urgent Federal Priority:** Multiple federal agencies and Congressional bodies have identified improved forecasting as a critical need, with Congress calling for "government action" and the federal commission emphasizing the need for *"modernizing tools for informed decision-making" (See Exhibits C3, C**2**, C7, C**1**)*.

4. **Growing Problem:** Climate change is projected to increase wildfire frequency, intensity, and smoke production, enlarging the already massive economic and health burdens *(See Exhibits C**2**, C**1**)*.

5. **Critical Technology Alignment:** The research directly advances federally designated Critical and Emerging Technologies (artificial intelligence and advanced computing) as identified by the National Science and Technology Council *(See Exhibit C6)*.

Taken together, these factors conclusively demonstrate that my proposed endeavor has profound national importance, satisfying the second prong of the Dhanasar framework.

Chapter Three

I Am Well-Positioned to Advance My Proposed Endeavor

The second prong of the NIW framework, as established in *Matter of Dhanasar*, requires demonstrating that I am well-positioned to advance my proposed endeavor. The decision identifies several factors that USCIS may consider, including:

1. the individual's education, skills, knowledge, and record of success in related or similar efforts;
2. a model or plan for future activities;
3. progress toward achieving the proposed endeavor; and
4. the interest of potential users, employers, or other relevant entities.

In *Dhanasar*, the petitioner satisfied this prong through advanced academic training, experience directly connected to the proposed endeavor, and significant contributions to government-funded research. Expert evaluations were also cited as important evidence.

My petition satisfies these same criteria. I present documentation of my academic qualifications, technical skills, record of success in data assimilation and wildfire smoke transport modeling, progress toward the proposed endeavor, and a clear plan for future activities. This evidence is supported by detailed expert letters *(SeeExhibits A1–A5)*, which affirm my exceptional capabilities and my strong position to advance research in model error estimation and data assimilation for wildfire smoke forecasting.

The following sections set out the evidence establishing that I am well-positioned to successfully advance this nationally important endeavor.

3.1 Academic Qualifications, Skills, and Knowledge

I am a computational scientist with advanced academic training, an exceptional academic record, and a highly specialized skill set that directly supports the proposed endeavor. My education and technical background position me to contribute at a high level to the development of improved model error estimation, data assimilation, and scientific machine learning tools for wildfire smoke forecasting.

Undergraduate Training (August 2016 – July 2019):

My academic foundation in mathematics and physics began at Busitema University in Uganda, where I earned a Bachelor of Science with Education in Physics and Mathematics (CGPA 4.20/5.0). This training provided rigorous preparation in mathematical analysis, classical and modern physics, and numerical reasoning. These skills form the basis for advanced work in atmospheric transport modeling, numerical simulation, and applied mathematics, which later became central to my proposed endeavor in wildfire smoke transport modeling. ***(See Exhibits B1, B2, B3)***

Master's-Level Advanced Mathematical Training (August 2020 – July 2021):

I strengthened this foundation through a Master of Science in Mathematical Sciences at the African Institute for Mathematical Sciences (AIMS) Rwanda ***(See Exhibits B4, B5, B6, B7, D22)***. I graduated with Distinction and I was on a fully funded scholarship. My master's thesis: "Assimilating Long-Range Regional Transport of Wildfire Smoke by Estimating $PM_{2.5}$ Concentration Using Weak Constraint Data Assimilation", directly addressed the scientific challenges that now define my proposed endeavor.

AIMS is internationally recognized for its highly selective admissions and exceptionally rigorous curriculum in applied mathematics, probability, statistics, numerical analysis, and scientific computing. Professor Jodi Mead confirms both the exceptional nature of this training and my performance:

"I first met Sandra at the African Institute for Mathematical Sciences (AIMS), a network of highly selective graduate centers across Africa known for their rigorous mathematical and computational training. Recognizing her exceptional ability, I recruited her to Boise State to pursue a Ph.D. in Computing." (See Exhibit A2)

This statement demonstrates that I was recruited into doctoral study on the basis of demonstrated mathematical and computational excellence, a key indicator that I am well-positioned to advance my proposed endeavor.

Doctoral Training and Research Excellence (January 2022 – Present):

I am currently pursuing a Ph.D. in Computing at Boise State University, specializing in Computational Mathematics, Science, and Engineering. I have completed all coursework with a summa cum laude CGPA of 3.968/4.0 ***(See Exhibits B8, B9, B10)***. My dissertation, "Advanced Computational Methods for Model Error Estimation in Variational Data Assimilation for Wildfire Smoke Transport Modeling," develops new mathematical and computational techniques for model error estimation in weak constraint variational data assimilation, specifically applied to wildfire smoke transport ***(See Exhibits D21)***.

My doctoral research has progressed through two interconnected phases that together position me to advance operational forecasting systems. The first phase, now published in the premier SIAM/ASA Journal on Uncertainty Quantification, established rigorous mathematical foundations for automated model error covariance estimation. The second phase extends these classical methods with physics-informed machine learning to achieve the computational efficiency required for real-time operational deployment.

a) Classical Data Assimilation Foundations

Professor Jodi Mead describes the technical significance of my published contributions:

"Sandra's doctoral research centers on four-dimensional variational data assimilation using the representer method… She has made notable contributions by reformulating model-error weighting as a regularization parameter selection problem, introducing new methodologies grounded in inverse theory and statistical optimization." (See Exhibit A2)

This published work addresses a fundamental challenge in operational forecasting: existing systems rely on ad-hoc tuning of model error parameters, while my automated framework provides mathematically rigorous, observation-informed covariance estimation. However, even with optimal parameter selection, classical iterative solvers (Gauss-Newton, MCMC) become computationally prohibitive for operational forecasting systems requiring rapid parameter updates during active wildfire events.

b) Extension to Physics-Informed Machine Learning

Building on this foundational work, my current research extends to physics-informed deep learning approaches that maintain the mathematical rigor of my published methods while achieving the computational efficiency federal agencies require. This progression directly addresses the operational bottleneck documented by the Government Accountability Office: current forecasting systems cannot fully utilize available satellite data because classical assimilation methods are too computationally intensive for real-time operational use *(See Exhibit C3, GAO Report 2023, p. 20).*

My machine learning research focuses on three interconnected frameworks:

- **Physics-Informed Neural Networks (PINNs):** I develop neural architectures that embed conservation laws and transport physics directly into loss functions, ensuring predictions respect fundamental atmospheric dynamics. By incorporating the advection-diffusion equation, atmospheric boundary conditions, and mass conservation constraints as soft penalties in the training objective, these networks achieve 10-100× computational speedup over classical methods while maintaining physical consistency. This approach transforms hours of iterative computation into seconds of neural network inference, a capability essential for the 6-12 hour forecast windows required for emergency evacuation planning and air quality alert systems.
- **Deep Operator Networks (DeepONets):** I train operator learning frameworks that map observation patterns to optimal emission estimates, enabling near-instantaneous parameter inference once trained. Unlike traditional methods that solve an inverse problem for each new observation set, trained DeepONets provide immediate predictions by learning the underlying operator mapping from sparse, noisy satellite observations to full three-dimensional emission fields. This capability is particularly critical for rapidly evolving smoke events where forecast updates must occur faster than real-time.
- **Hybrid Classical-ML Frameworks:** I combine the uncertainty quantification rigor of my published weak-constraint 4D-Var methods with ML computational efficiency. This

> hybrid approach uses classical methods to generate high-fidelity training data and validate neural network predictions, while deploying trained networks for operational real-time forecasting. The resulting forecasts maintain statistical reliability and quantified uncertainty bounds while meeting the operational time constraints identified by federal agencies including NOAA, NASA, and the Naval Research Laboratory.

This dual expertise in both classical data assimilation and scientific machine learning positions me uniquely to advance operational forecasting systems. As federal researchers have documented, purely ML-trained researchers often lack the domain knowledge to embed physical constraints correctly, while traditional data assimilation researchers may lack the implementation skills for production-ready neural architectures. My research explicitly bridges this gap by combining deep mathematical foundations in inverse problems and uncertainty quantification with cutting-edge machine learning techniques designed specifically for operational deployment.

Professor Mead confirms my exceptional standing: ***"Based on my experience advising numerous doctoral researchers, I can state without reservation that Sandra Babyale ranks among the very best. Her combination of deep mathematical insight, computational skill, and perseverance in solving complex problems is exceptional." (See Exhibit A2)***

These evaluations, combined with my published contributions and ongoing research, demonstrate that I possess the advanced training, intellectual ability, and technical expertise necessary to advance the proposed endeavor at a national level.

Technical Skills and Competencies

My technical expertise spans numerical modeling, data assimilation, inverse problems, scientific machine learning, and high-performance computing. I am proficient in finite difference and finite volume methods, adaptive mesh refinement, weak constraint 4D-Var, representer methods, and ensemble data assimilation techniques. My work incorporates advanced inverse problem theory, uncertainty quantification, and model error estimation, all essential components of modern atmospheric forecasting systems.

In addition, I have extensive experience implementing scientific machine learning methods, including deep neural networks and physics-informed learning frameworks. My computational work is supported by strong high-performance computing skills, including MPI, OpenMP, CUDA, and parallel computation on large HPC clusters. I am proficient in multiple programming languages (Python, R, MATLAB, C++, C, FORTRAN, Java) and scientific tools such as Git/GitHub, NCL, Ferret, VS Code, Bash scripting, and LaTeX.

These skills are central to the development of modern data assimilation and environmental forecasting systems used by federal agencies such as NOAA, NASA, the EPA, and the U.S. Forest Service. Combined with the expert assessments above, they demonstrate that I possess the advanced academic preparation, technical capability, and computational expertise required to successfully advance my proposed endeavor.

My sustained engagement with the applied mathematics and computational science community is reflected in my long-standing membership in the Society for Industrial and Applied Mathematics (SIAM) since 2022. SIAM is the primary professional society for researchers working in inverse problems, uncertainty quantification, numerical analysis, and data assimilation, the core methodological foundations of my research. This professional affiliation situates my work within the central disciplinary community advancing mathematical tools used in national environmental prediction systems *(See Exhibit D17)*.

3.2 Record of Success in Related Efforts and Progress Toward the Proposed Endeavor

My research in the United States has directly contributed to nationally important priorities in wildfire smoke forecasting, public health protection, and climate resilience. I have established a strong record of success through federally funded research, peer-reviewed publications, conference presentations, software development, and collaborations with senior experts in data assimilation and atmospheric modeling. These accomplishments demonstrate measurable progress toward my proposed endeavor and confirm that I possess the experience, technical capability, and research trajectory required to advance this work at a national level.

3.2.1 NSF-Funded Research on Wildfire Smoke Transport

I am a primary doctoral researcher on the National Science Foundation funded project "Data-Enabled Modeling of Wildfire Smoke Transport" (NSF-DMS Award #2111585, nearly $550,000 over three years) *(See Exhibit D3).* My research directly supports the project's scientific objectives by developing the mathematical and computational methods that enable accurate wildfire smoke forecasting with quantified uncertainty. This federally funded project addresses one of the most urgent national challenges: developing accurate and computationally efficient techniques to predict wildfire smoke transport, assess forecast reliability, and improve public health relevant air quality prediction systems.

The national significance of this NSF investment is documented publicly. As reported by Boise State University News on October 19, 2021:

"Mathematics professor Jodi Mead and associate professor Donna Calhoun recently received a National Science Foundation grant of nearly $550,000 over three years in order to computer model the transport of wildfire smoke. The goal of this work is to develop techniques to predict the smoke forecast in a given region, and then assess the accuracy of those computer modeled predictions." (See Exhibit D3)

The article identifies me as one of the core researchers working directly with Professors Mead and Calhoun on this federally funded project. It further quotes Principal Investigator Professor Mead, who emphasizes the practical difficulty the project is designed to solve:

"On the news now, just as they give the temperature and precipitation, they give a smoke forecast. But the issue that we're trying to address is how accurate the forecast is because when you get into more rural areas, we don't have as many observing systems." (See Exhibit D3)

This statement directly reinforces the central problem addressed by my proposed NIW endeavor: improving accuracy and uncertainty quantification in smoke forecasting systems through advanced data assimilation and model error estimation techniques.

My Research Contributions to the NSF Project:

- **Mathematical Framework Development**: Formulating model error covariance estimation as a regularization parameter selection problem, introducing rigorous inverse theory based methods to replace ad-hoc tuning approaches currently used in operational forecasting systems.
- **Algorithm Design and Validation**: Developing and testing weak constraint four-dimensional variational (4D-Var) data assimilation algorithms using representer methods, enabling the atmospheric transport model to optimally combine observational data with physical model predictions while accounting for model structural uncertainties.
- **Numerical Experimentation**: Designing and conducting systematic numerical experiments using one-dimensional, two-dimensional, and three-dimensional advection-transport models to validate model error estimation techniques under varying atmospheric conditions and observational coverage scenarios.
- **Scientific Machine Learning Integration**: Developing neural network architectures (including Physics-Informed Neural Networks and Deep Operator Networks) for solving nonlinear geophysical inverse problems, enabling efficient estimation of uncertain emissions, boundary conditions, and model parameters from sparse observational data.
- **High-Performance Computing Implementation**: Creating GPU-accelerated transport solvers using CUDA programming, achieving significant computational speedup that enables real-time data assimilation for operational forecasting applications.
- **Computational Infrastructure Development**: Parallelizing large-scale atmospheric simulations using MPI and OpenMP on Borah, Boise State University's supercomputing cluster, enabling ensemble-based uncertainty quantification across multiple model realizations.
- **Observational Data Integration**: Developing methods to assimilate satellite aerosol optical depth (AOD) measurements, ground-based $PM_{2.5}$ station data, and fire radiative power observations into the atmospheric transport framework, directly addressing federal findings that satellite data remains significantly underutilized in current operational systems *(See Exhibit C3, GAO-24-106213, p. 20)*.

These contributions directly fulfill NSF’s mandate to advance the scientific foundations of wildfire smoke modeling and address documented gaps in national forecasting capabilities. My research provides the mathematical rigor, computational efficiency, and uncertainty quantification methods that distinguish advanced research models from operational systems currently limited by incomplete model error treatment.

Principal Investigator Confirmation

Professor Donna Calhoun, co-Principal Investigator of the NSF project and an expert with extensive federal funding from NSF, DARPA, NASA, and DOE, provides authoritative confirmation of my essential contributions:

"Sandra's research focuses on developing a framework for modeling covariance estimation in weak-constraint variational data assimilation. This work is crucial to addressing key uncertainties in our computational smoke model, Smoke3d. As corresponding author on our recent joint paper "Model Covariance Estimation for Weak Constraint Variational Data Assimilation" and responsible for much of the writing and most of the simulation results in that paper, Sandra has impressed me with how she has developed into a focused and serious applied mathematician." (See Exhibit A3)

Her testimony confirms that my research contributions are scientifically indispensable to this major federally funded initiative and that I have produced original methodological advances recognized by senior experts in the field.

Summary

This NSF-funded project demonstrates:

- Federal recognition of the national importance of wildfire smoke forecasting and the need for advanced mathematical and computational methods,
- My proven ability to contribute meaningfully to federally prioritized research areas at the doctoral level,
- Direct evidence that my research addresses documented scientific gaps in national forecasting systems identified by federal agencies including GAO, NOAA, NASA, and EPA,
- Ongoing progress toward the proposed NIW endeavor through research that is already supported by substantial federal investment.

Together, the federal documentation, published research outcomes, and expert evaluations provide strong evidence that I have already made substantial scientific progress toward the proposed endeavor and am well-positioned to advance it at the national level through continued research, publication, and collaboration with federal agencies and research institutions.

3.2.2 Peer-Reviewed Journal Publications

Published/Accepted:

1. **Sandra R. Babyale**, Jodi Mead, Donna Calhoun, Patricia Azike. "Model Error Variance Estimation for Weak-Constraint Data Assimilation." *SIAM/ASA Journal on Uncertainty Quantification* (Published, 2025). *(See Exhibit D1)*

This publication presents a new mathematical framework for estimating model error variance in weak-constraint data assimilation applied to wildfire smoke transport. The work incorporates regularization parameter selection methods, L-curve and generalized cross-validation analysis, and representer-based formulations. As corresponding author, I led the writing of the manuscript and was responsible for the majority of the simulation results and theoretical development.

The SIAM/ASA Journal on Uncertainty Quantification is one of the most selective and prestigious journals in applied mathematics and computational science, jointly published by the Society for Industrial and Applied Mathematics (SIAM) and the American Statistical Association (ASA). The journal maintains rigorous peer-review standards and focuses on high-impact research in uncertainty quantification, inverse problems, and data assimilation, areas of critical importance to federal environmental prediction systems.

Significance of Publication Venue

The acceptance of this work in SIAM/ASA JUQ is particularly meaningful given the national shortage of researchers in variational data assimilation documented by federal agencies. As Dr. Hans Ngodock of the U.S. Naval Research Laboratory notes: ***"It is very difficult to train or to find scientists in variational data assimilation. One of the reasons being that many pioneers in this field have retired, and most departments in the academia prefer teaching and using simpler methods for data assimilation. " (See Exhibit A1)***

The publication of advanced methods in this area addresses a documented federal need. The U.S. Government Accountability Office specifically identified model error representation and data assimilation as limiting factors in current wildfire smoke forecasting systems, stating that *"satellite data is significantly underutilized"* due to inadequate assimilation methods *(See Exhibit C3, GAO-24-106213, p. 20).* My published research directly provides mathematical solutions to this federally identified problem.

Expert Recognition of Research Contribution

Dr. Mojtaba Sadegh writes: ***"Her manuscript on model error estimation was published in the SIAM ASA Journal on Uncertainty Quantification, a leading journal in applied mathematics and computational science. Publication in this journal confirms that her work has undergone rigorous peer review and has been recognized by experts in the field." (See Exhibit A4)***

Dr. Hans Ngodock of the U.S. Naval Research Laboratory states: ***"I have read both Sandra's resume and her manuscript submitted for publication in the world-renown SIAM/ASA Journal on Uncertainty Quantification. This manuscript attests not only to Sandra's understanding of variational methods and applications to data assimilation in general, but also to her innovative mind and contribution to advancing the scientific knowledge in those areas." (See Exhibit A1)***

These assessments demonstrate that my research contributions are both technically advanced and nationally relevant to environmental prediction. Published in 2025, this work introduces novel methodologies that are now available to the atmospheric modeling and data assimilation communities. The acceptance and publication following rigorous peer review by leading experts confirms the scientific validity and importance of these contributions. As this work addresses documented federal needs in wildfire smoke forecasting and model error estimation, it is positioned to influence ongoing research at agencies such as NOAA, NASA, EPA, and the Naval Research Laboratory.

In Preparation:

2. *"Deep Neural Networks for Nonlinear Geophysical Inverse Problems: Sampling, Convergence, and Applications to Wildfire Smoke Data Assimilation."* (Author order: Babyale, Mead, Calhoun, Azike)

3. *"Data-Enabled Modeling of Wildfire Smoke Transport: Smoke3d."* (Author order: Calhoun, Mead, Babyale, Azike)

3.2.3 Peer-Reviewed Conference Presentations

I have presented my research at major national and international conferences in applied mathematics, atmospheric science, and wildfire modeling. All presentations were peer-reviewed for acceptance.

Oral Presentations:

1. ***"Model Error Covariance Estimation for Weak Constraint Variational Data Assimilation."*** SIAM Pacific Northwest Section Meeting, University of Washington, Seattle, WA (October 2025) SIAM PNW 2025 *(See Exhibits D5, D7)*

2. ***"Choice of Model Error Variance for Data Assimilation with the Transport Equation."*** SIAM Pacific Northwest Section Meeting, Western Washington University, Bellingham, WA (October 2023) SIAM PNW 2023 *(See Exhibits D4, D6)*

Poster Presentations:

3. ***"Model Error Covariance Estimation for Weak Constraint Data Assimilation Using Regularization Parameter Selection Methods."*** IMSI Workshop on Uncertainty Quantification and Machine Learning for Complex Physical Systems, University of Chicago, IL (2025). Funded by NSF Institute for Mathematical and Statistical Innovation. *(See Exhibit D14)*

4. ***"Model Variance Estimation for Variational Data Assimilation Using Regularization Parameter Selection Methods."*** AWM Workshop, SIAM Annual Meeting, Spokane, WA (July 2024). AWM Workshop**. Recipient of the AWM Poster Award.** *(See Exhibit D9)*

5. ***"Data-enabled Modeling of Wildfire Smoke Transport."*** 7th International Fire Behavior and Fuels Conference, Boise, ID (April 2024). *(See Exhibit D16)*

6. ***"Data-enabled Modeling of Wildfire Smoke Transport."*** Graduate Student Showcase, Boise State University (2023). *(See Exhibit D11)*

7. **Co-Authored Presentation: Donna Calhoun, Sandra R. Babyale, Jodi Mead.** *"A Data-Enabled Wildfire Smoke Transport Model."* SIAM Pacific Northwest Section Meeting, Vancouver, WA (2022) Link

My research is informed by direct engagement with the wildfire science and practitioner community through my active membership in the International Association of Wildland Fire (IAWF). As part of this engagement, I participated in the 7th International Fire Behavior and Fuels Conference, a leading international forum where researchers, operational scientists, and land management professionals address real-world wildfire behavior, smoke transport, and mitigation challenges. This involvement ensures that my computational and mathematical research remains closely aligned with operational needs in wildfire smoke forecasting and public health protection. *(See Exhibit D16)*

Dr. Mojtaba Sadegh states: ***"she attended the 7th International Fire Behavior and Fuels Conference, one of the primary meetings for the global fire science community. I presented research at that same conference. Her participation indicates that she is already engaging with researchers and practitioners working on (inter)national wildfire challenges." (Exhibit A4)***

3.2.4 Software Contributions

Smoke3D (https://github.com/ForestClaw/Smoke3d)

I am among the primary contributors to Smoke3d, a high-performance wildfire smoke transport modeling system created under the NSF-funded project *"Data-Enabled Modeling of Wildfire Smoke Transport."* My research contributions to this software include:

- Development and validation of model error estimation algorithms

- Design and implementation of weak-constraint data assimilation methods
- Numerical experimentation and testing in 1D, 2D, and 3D transport scenarios
- Development of GPU-accelerated transport solvers using CUDA
- Performance benchmarking and optimization on Boise State's Borah supercomputer
- Integration of regularization-based covariance estimation techniques

Smoke3D utilizes adaptive mesh refinement and efficient numerical methods to support hazard preparedness, air-quality monitoring, and emergency management. It will be released as open-source software through the ForestClaw ecosystem, enabling widespread use by federal agencies and the research community.

Upon completion, Smoke3d will be a standalone open-source software package linked to the ForestClaw framework (https://github.com/ForestClaw/forestclaw), expanding accessibility for the scientific community and government agencies involved in wildfire and atmospheric hazard prediction. Its open-source release will enable NOAA, EPA, NASA, and academic researchers to integrate improved model error estimation and data assimilation methods directly into operational and research workflows, increasing the national impact of my contributions.

3.2.6 Summary of Scholarly Contributions

Collectively, my scholarly work demonstrates a sustained record of success in the scientific and computational areas central to my proposed endeavor. Through peer-reviewed publications, including a paper accepted in the SIAM/ASA Journal on Uncertainty Quantification, I have advanced new mathematical and computational methods for model error estimation in weak-constraint variational data assimilation. My conference presentations at SIAM, IMSI, and the International Fire Behavior and Fuels Conference further show that my work has been vetted by experts and disseminated within leading scientific communities.

My contributions to large-scale modeling software, including my role as a core developer of Smoke3d, provide practical tools that directly support wildfire smoke prediction research funded by the National Science Foundation. These software developments, combined with my research in machine learning and inverse problems, demonstrate an ability to translate mathematical theory into high-performance computational systems used by federal agencies and the broader scientific community.

Taken together, these outputs reflect a strong publication record, active scientific engagement, and impactful technical contributions that position me to advance the proposed endeavor at a national level.

3.2.6 Honors and Awards

I have received multiple competitive awards at the university, national, and international levels in recognition of my academic excellence, research contributions, and potential for leadership in computational mathematics and environmental modeling. These distinctions demonstrate that experts and scientific institutions have independently evaluated my abilities and selected me for opportunities based on merit, an important indicator that I am well-positioned to advance my proposed endeavor.

1. **NSF-Funded Graduate Research Support & Graduate Assistantship Awards (2022–present)**

From 2022 to 2025, I was fully funded through a National Science Foundation (NSF) research grant supporting the project *"Data-Enabled Modeling of Wildfire Smoke Transport."* This federal support covered my doctoral research, tuition, stipend, and health insurance. My continued performance and contributions led the Ph.D. in Computing Program at Boise State University to offer me a competitive twelve-month Graduate Assistantship for the 2025-2026 academic year, providing a stipend, full tuition, and health insurance.

This continuous sequence of federally supported and merit-based institutional funding reflects sustained recognition of my academic excellence, research productivity, and value to the university's federally funded wildfire smoke modeling program. *(See Exhibit D3, B8, B10)*

2. **Poster Award, Association for Women in Mathematics (AWM) Workshop, SIAM Annual Meeting, Spokane, WA (2024)**

My poster *"Model Variance Estimation for Variational Data Assimilation Using Regularization Parameter Selection Methods"* received a competitive award at one of the largest applied mathematics conferences in the world. This honor recognizes the quality and impact of my research in data assimilation and inverse problems. *(See Exhibit D9)*

3. **Competitive Stipend Award, IMSI Data Science Summer Bootcamp, University of Chicago (2025)**

I was selected as a semifinalist for the competitive IMSI Internship Program and, based on merit, was invited to participate in the fully funded two-week IMSI Data Science Summer Bootcamp (June 2025). The Institute provided a $1,400 stipend for participation, and the program delivered advanced training in applied machine learning, scientific computing, and modern data-science workflows. IMSI is an NSF-funded Mathematical Sciences Research Institute, and participation is limited to selected applicants. *(See Exhibit D13)*

4. **Travel Award, Institute for Pure and Applied Mathematics (IPAM) Workshop, UCLA (2025)**

I was selected for competitive funding by IPAM, a National Science Foundation Mathematical Sciences Institute known for its rigorous and highly selective research programs. This award signifies national recognition of my expertise in computational mathematics and scientific machine learning. *(See Exhibit D15)*

5. **Travel Award, Institute for Mathematical and Statistical Innovation (IMSI) Workshop, University of Chicago (2025)**

I received competitive funding to present my poster at the NSF IMSI Workshop on Uncertainty Quantification and Machine Learning for Complex Physical Systems. This selection reflects recognition from a premier institute at the forefront of mathematical and statistical innovation. *(See Exhibit D14)*

6. **Travel Award, U.S. Research Software Sustainability Institute (US-RSSI) Summer School, George Washington University (2024)**

I was competitively selected to participate in and funded to attend intensive national training in software engineering, high-performance computing, and research software design, demonstrating my contributions to computational infrastructure and scientific software. *(See Exhibit D12)*

7. **Travel Award, AWM Workshop at SIAM Annual Meeting, Spokane, WA (2024)**

This highly competitive AWM award supported my participation at SIAM AN24, reinforcing both the novelty of my research and my potential as a future leader in applied mathematics. *(See Exhibit D9)*

8. **First Place Winner, BSU Research Computing Days SIAM Hackathon, Boise State University (2023)**

My team won first place in a university-wide computational hackathon hosted by the Boise State SIAM chapter, reflecting my strong problem-solving skills, programming expertise, and ability to innovate under time constraints. *(See Exhibit D10)*

9. **Fully Funded Master's Scholarship, African Institute for Mathematical Sciences, Rwanda (2020–2021)**

I was competitively selected for a fully funded scholarship at AIMS, an internationally recognized center of excellence in mathematical sciences. This award reflects exceptional academic promise and performance at the graduate level. *(See Exhibit B4)*

10. **Undergraduate Government Scholarship, Busitema University, Uganda (2016–2019)**

I received a full government scholarship for undergraduate study based on academic merit, demonstrating early recognition of my capabilities in mathematics and science.

Professor Jodi Mead notes: ***"Beyond her technical contributions, Sandra has been recognized for excellence and leadership. She presented award-winning research at the SIAM Annual Meeting (2024) and actively engages with national professional networks, including the U.S. Research Software Sustainability Institute and the NSF Institute for Mathematical and Statistical Innovation. These activities demonstrate her readiness to contribute meaningfully to the nation's scientific infrastructure and workforce." (See Exhibit A2)***

Collectively, these awards and recognitions demonstrate sustained excellence, competitive selection at national and international levels, and engagement in scientific communities directly aligned with the proposed endeavor.

3.3 Teaching, Leadership, and STEM Outreach

I have demonstrated strong teaching ability, leadership in the applied mathematics community, and sustained commitment to STEM outreach. These activities reflect both my pedagogical competence and my dedication to strengthening the U.S. STEM workforce pipeline.

1. **Instructor of Record, MATH 170 (Calculus I), Boise State University (August 2025 – Present)**

I serve as the instructor of record for Calculus I, one of the most critical gateway courses for students pursuing careers in engineering, science, technology, and other quantitative fields. In this role, I design course materials, lead lectures, mentor students, and evaluate their mastery of foundational mathematical concepts.

Professor Joe Champion, a tenured Professor of Mathematics with more than 23 years of experience in STEM education, directly supervises my teaching. He notes:

"Due to her exceptional qualifications, we entrusted Ms. Babyale with the responsibility of teaching Calculus I as instructor of record during the Fall 2025 semester, and I have supervised her performance. Immediately, Ms. Babyale has distinguished herself through her extraordinary initiative. Prior to the start of her teaching assignment, she proactively sought my mentorship and voluntarily observed my own classes to learn and adopt effective pedagogical strategies." (See Exhibit A5)

Professor Champion further emphasizes the national impact of this work:

"Calculus I is a crucial gateway course that influences the progress of many aspiring engineers, scientists, and technology professionals. It is also a well-known attrition point where many students, particularly from underrepresented backgrounds, abandon STEM pathways. Retaining these students is a matter of national strategic importance ... She is not

just teaching a class; she is actively strengthening the U.S. STEM pipeline and ensuring the next generation of U.S. STEM innovators have the knowledge to succeed." (See Exhibit A5)

These assessments confirm both the trust placed in me as an educator and the national importance of my instructional contributions.

2. **President, SIAM Student Chapter, Boise State University (August 2025 – Present)**

As president of the SIAM Student Chapter, I lead academic and professional development initiatives that promote applied mathematics and computational science. I organize outreach events at local high schools, coordinate seminars, and mentor undergraduate students interested in research careers.

Professor Champion remarks on the importance of this leadership: ***"In addition, she has a leadership role in the Boise State Chapter of the Society for Industrial and Applied Mathematics (SIAM), where she has organized outreach events to encourage local high school students to pursue careers in mathematics."(See Exhibit A5)***

This role demonstrates my capacity to contribute to national workforce development through mentorship and community engagement.

3. **Volunteer, Boise State Math Invitational (2022 – Present)**

I volunteer annually at the Boise State Math Invitational, a statewide competition that promotes mathematical problem-solving among high school students. My involvement includes mentoring teams, supporting event logistics, and encouraging the next generation of STEM learners.

Professor Champion highlights the significance of this outreach: ***"Ms. Babyale's commitment to advancing STEM education extends far beyond her required duties. She has delivered volunteer service for the Boise State's High School Math Invitational, an important local event that connects the university with high school students across the region." (See Exhibit A5)***

Through university-level instruction, leadership in professional organizations, and ongoing STEM outreach, I contribute meaningfully to strengthening the United States STEM talent pipeline. These activities are consistent with national priorities to expand STEM participation and ensure a well-prepared scientific workforce. Combined with my research accomplishments, they further demonstrate that I am well-positioned to advance my proposed endeavor.

3.4 Professional Development and Training

I actively pursue advanced professional training opportunities that strengthen my ability to contribute to national research priorities in data assimilation, machine learning, scientific computing, and wildfire modeling.

1. **IMSI Summer Data Science Bootcamp, University of Chicago (June 2025) -** An intensive two-week training focused on Python and R for numerical analysis, image processing, and text analytics. Participants receive supported instruction from leading data science faculty. *(See Exhibit D13)*

2. **U.S. Research Software Sustainability Institute (US-RSSI) Summer School, George Washington University (June 2024) -** Training in software design, modular programming, version control (Git/GitHub), automated testing, packaging, and scientific reproducibility. The program emphasizes open science and best practices for building research-grade scientific software. *(See Exhibit D12)*

3. **IPAM Workshop on Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning, UCLA (2025) -** A selective workshop providing advanced training in machine learning for physical systems, including sampling methods, inference, operator learning, and physics-informed modeling. *(See Exhibit D15)*

4. **LeaderShape Institute, Boise State University (May 2024) -** A competitive leadership program focused on vision-driven leadership, effective communication, and collaborative problem-solving.

Professor Donna Calhoun highlights the significance of my professional development:

"Sandra has impressed me with how she has developed into a focused and serious applied mathematician. Sandra has also impressed me with her willingness to seek out opportunities to broaden her research experience. Recently, she applied and was accepted to an IPAM

(UCLA) workshop on machine learning. This workshop was particularly relevant to her goal to pursue research in machine learning for scientific applications. Machine learning and more generally, artificial intelligence are areas where the US will need top talented individuals from all parts of the world. Sandra's interest and skills in research in this area will make her highly sought after by employers in academia, high-tech industries, and other research institutions. And her particular interest in wildfire smoke modeling is becoming increasingly relevant." (See Exhibit A3)

Professor Calhoun's assessment of my machine learning qualifications is particularly significant given the national workforce shortage in this area. Her identification of artificial intelligence and machine learning as fields ***"where the US will need top talented individuals from all parts of the world" (See Exhibit A3)*** directly aligns with the National Science and Technology Council's designation of AI, machine learning, advanced computing, advanced modeling and simulation as Critical and Emerging Technologies essential to U.S. national security *(See Exhibit C6, p. 2, 3, 4)*.

My strategic pursuit of advanced ML training at NSF-funded institutes (IPAM, IMSI) combined with my deep foundation in classical data assimilation creates the exact dual expertise federal agencies require but struggle to find: the domain knowledge to embed physical constraints correctly paired with the implementation skills to develop production-ready neural architectures.

3.5 Professional Memberships

I maintain active membership in leading professional organizations relevant to wildfire science, atmospheric modeling, and applied mathematics. I am a member of the International Association of Wildland Fire (IAWF), which connects researchers and practitioners addressing wildfire behavior and smoke impacts. I am also a member of the Society for Industrial and Applied Mathematics (SIAM), the principal professional society for applied and computational mathematicians working in data assimilation and inverse problems. In addition, I am a member of the American Geophysical Union (AGU), reflecting the interdisciplinary nature of my work at the interface of applied mathematics, atmospheric science, and environmental prediction *(See Exhibit D18)*.

3.6 Progress Toward Achieving the Proposed Endeavor and Plan for Future Activities

3.6.1 Progress Towards Achieving the Proposed Endeavor

I have already made substantial progress in advancing the proposed endeavor of developing improved data assimilation and machine learning techniques for wildfire smoke transport forecasting. Key accomplishments include:

Peer-Reviewed Publication - Lead-author paper published in SIAM/ASA Journal on Uncertainty Quantification (2025), introducing a new automated framework for estimating model error covariance in weak-constraint 4D-Var. This work establishes a principled approach for balancing model predictions with observational data and removes reliance on ad-hoc tuning.

NSF-Funded Research Contributions - Primary researcher on NSF-DMS project "Data-Enabled Modeling of Wildfire Smoke Transport," contributing directly to the development of Smoke3d, a high-performance computational model for wildfire smoke transport used to evaluate new model error estimation techniques.

Conference Dissemination - Seven peer-reviewed presentations at national and international conferences, including SIAM, AWM, and IMSI, demonstrating active and ongoing dissemination of research findings relevant to wildfire smoke forecasting, data assimilation, and uncertainty quantification.

Award Recognition - Recipient of the Poster Award at the 2024 AWM Workshop during the SIAM Annual Meeting, recognizing excellence in research related to model error estimation and variational data assimilation.

Software Development - I am among the developers of Smoke3d, an open-source, data-enabled computational model for wildfire smoke transport that will support air-quality forecasting, hazard readiness, and environmental prediction for agencies such as NOAA, EPA, and the U.S. Forest Service.

Collectively, these achievements demonstrate a sustained, concrete, and accelerating trajectory toward advancing the proposed endeavor.

3.6.2 Plan for Future Activities

I am fully committed to continuing and expanding my research in wildfire smoke forecasting in the United States. My forward plan is structured in short-, medium-, and long-term phases that align with national needs in environmental prediction, model error estimation, and scientific machine learning.

Short-Term (1-2 years):

- Complete Ph.D. dissertation on model error estimation and data-enabled wildfire smoke transport.
- Publish additional papers on neural inverse problem solvers, model error learning, and hybrid data assimilation methods.
- Finalize Smoke3d as a standalone open-source software package integrated with ForestClaw for broad governmental and research use.
- Continue developing efficient sampling strategies and neural architectures (FNNs, CNNs, DeepONets, PINNs) for ill-posed inverse problems directly relevant to data assimilation.
- Seek research scientist or postdoctoral positions at U.S. universities, national laboratories, or federal agencies.

Medium-Term (3-5 years):

- Integrate real-time satellite data into Smoke3D and expand weak-constraint 4D-Var capabilities.
- Develop neural operators capable of learning dynamic model-error structure from evolving atmospheric conditions.
- Advance hybrid data assimilation systems that blend classical covariance models with neural operator architectures to reduce computational cost and improve forecast skill.
- Establish collaborations with NOAA, NASA, EPA, and national labs to transition research innovations into pre-operational forecasting tools.

- Pursue independent federal research funding through NSF, NOAA, NASA, and DOE.

Long-Term (5+ years):

- Lead a U.S.-based research program focused on next-generation wildfire smoke forecasting systems, combining mathematical modeling, machine learning, and operational data assimilation.
- Train and mentor students and junior researchers in this nationally critical area.
- Contribute to national standards for model error estimation, uncertainty quantification, and data assimilation for environmental hazards.
- Support federal efforts to safeguard public health, improve climate resilience, and enhance operational environmental prediction systems.

3.7 Interest of Potential Users, Employers, and Federal Agencies

Federal agencies, national laboratories, and leading researchers have independently expressed interest in my expertise, confirming that I am well-positioned to advance my proposed endeavor. My work aligns directly with ongoing national needs in wildfire emissions modeling, atmospheric transport, data assimilation, and scientific machine learning, areas where federal agencies require additional highly trained specialists.

3.7.1 National Science Foundation

My doctoral research is funded by the National Science Foundation (Award #2111585, $550,000). NSF support reflects both the national importance of the scientific problem and the confidence placed in my ability to contribute to advancing wildfire smoke transport modeling. My work on this project includes model development, numerical simulation, data assimilation, and machine-learning methods that directly support NSF's strategic goals in environmental prediction and computational science.

3.7.2 NASA Earth Exchange (NEX) and NASA Postdoctoral Program Interest

NASA scientists have recently identified my research as directly relevant to their work on wildfire emissions, air quality modeling, and data assimilation. On December 2, 2025, Dr. Arthur Mizzi of the NASA Earth Exchange (NEX) at Ames Research Center contacted me to initiate a meeting and evaluate my suitability for a NASA Postdoctoral Program (NPP) position. He wrote:

"You've been doing some very interesting work. We'd like to set up a meeting with you… The purpose of this meeting is to see if your work/skill set is suitable for pursuing a NASA Postdoctoral Program position. We can discuss what that means/entails as well as what we do in the NASA Earth Exchange (NEX) group." (See Exhibit 19, Dr. Arthur P. Mizzi, NASA Earth Exchange)

NASA's outreach demonstrates that my expertise in wildfire emissions, data assimilation, and air-quality modeling is aligned with mission-critical research conducted at NASA NEX. This unsolicited interest provides strong evidence that I am well-positioned to contribute to national research priorities and federal environmental forecasting systems.

3.7.3 U.S. Naval Research Laboratory (NRL)

The U.S. Naval Research Laboratory, a leading federal institution in operational data assimilation, has expressed direct interest in recruiting me. Dr. Hans Ngodock states: ***"As one that has recruited several PhD graduates over the years and mentored them into successful scientific careers of their own in data assimilation using variational methods, I will not hesitate to recruit Sandra upon her graduation. She will make significant contributions to our modeling and assimilation research efforts in support of the US Navy." (See Exhibit A1)***

This testimony provides clear evidence that a major federal research institution recognizes the value and relevance of my expertise.

3.7.4 National Workforce Need and Shortage of Experts

Multiple experts corroborate that the United States faces a critical shortage of specialists in data assimilation, model error estimation, wildfire smoke modeling, and scientific machine learning, precisely the domains of my proposed endeavor.There is a significant national shortage of researchers with my specialized expertise.

Dr. Mojtaba Sadegh confirms: ***"There is a significant shortage of researchers with expertise in wildfire smoke transport modeling, advanced data assimilation, model error estimation, and scientific machine learning for environmental systems. Ms. Babyale's skill set directly fills this national workforce gap and supports ongoing federal efforts to strengthen environmental prediction and climate resilience." (See Exhibit A4)***

Dr. Hans Ngodock highlights the scarcity of experts in my field: ***"It is very difficult to train or to find scientists in variational data assimilation. One of the reasons being that many pioneers in this field have retired, and most departments in the academia prefer teaching and using simpler methods for data assimilation." (See Exhibit A1)***

Professor Donna Calhoun similarly emphasizes national demand for my expertise: ***"Sandra Babyale's expertise, qualifications and career interests are extremely relevant to missions of many science related agencies, and also highly sought after by these agencies. The US has long had to seek expertise in computational mathematics from abroad, exactly because these positions are difficult to fill with US citizens." (See Exhibit A3)***

These assessments show that my research addresses a scientifically complex area where the U.S. workforce is insufficient to meet national needs.

3.7.5 Summary

The independent interest of NASA, the NSF, the U.S. Naval Research Laboratory, and leading domain experts provides powerful evidence that:

- Federal agencies recognize the value and applicability of my expertise
- My skills are directly aligned with national missions in wildfire smoke prediction and environmental modeling
- There is strong national demand and a workforce shortage in exactly the areas my work advances

Taken together, these factors demonstrate that I am exceptionally well-positioned to advance my proposed endeavor in the United States.

3.8 Expert Opinions on My Qualifications

Independent expert testimony from senior scientists and professors confirms that I possess the advanced technical expertise, research capability, and professional readiness required to advance my proposed endeavor at a national level. These experts collectively highlight my exceptional mathematical and computational ability, my rare skill set in variational data assimilation and machine learning, and my strong trajectory as a researcher.

Dr. Hans Ngodock, senior Oceanographer at the U.S. Naval Research Laboratory and one of the nation's foremost experts in variational data assimilation, emphasizes both my technical skill and scientific innovation: ***"This manuscript attests not only to Sandra's understanding of variational methods and applications to data assimilation in general, but also to her innovative mind and contribution to advancing the scientific knowledge in those areas." (See Exhibit A1)***

He further underscores the national relevance of my skill set: ***"It is very difficult to train or to find scientists in variational data assimilation … Sandra's work and contributions are applicable to the scientific research being conducted in many laboratories and national agencies, … making her a highly valuable asset for the US." (See Exhibit A1)***

Professor Donna Calhoun, an NSF- and DOE-funded computational mathematician with extensive experience evaluating federal research proposals, provides a strong assessment of my research ability and professional promise: ***"She is an extremely capable and serious researcher and already demonstrates an eagerness to take on challenging projects. She has excellent written and communication skills, and will be an asset to any research oriented institution." (See Exhibit A)***

She also affirms my overall qualifications: ***"Based on Sandra's commitment to advancing her career in the US, her exceptional talent and experience in both applied and computational mathematics, her integrity, her resourcefulness and her capacity for creative independent thought, I recommend Sandra without hesitation for approval of the permanent residence petition." (See Exhibit A3)***

Professor Joe Champion, who evaluates my teaching and leadership contributions, confirms the breadth of my professional strengths: ***“Ms. Babyale’s combination of advanced research, in-class teaching excellence, and external community outreach is rare and necessary. She is not only a promising researcher in computational mathematics but also a proven and passionate STEM educator.” (See Exhibit A5)***

Dr. Mojtaba Sadegh, a leading researcher in wildfire science and climate analytics, confirms that my technical expertise aligns directly with national scientific priorities: ***“Her combined expertise in data assimilation, inverse problems, and machine learning aligns with national research priorities in artificial intelligence for environmental and climate applications.” (See Exhibit A4)***

Collectively, these expert assessments demonstrate that I possess the advanced education, technical capability, research productivity, and professional readiness required to advance my proposed endeavor. Their evaluations provide strong, independent confirmation that I am well-positioned under the second prong of the Dhanasar framework.

3.9 Conclusion of Chapter 3: I Am Well-Positioned to Advance My Proposed Endeavor

The evidence presented in this chapter demonstrates that I am exceptionally well-positioned to advance my proposed endeavor of developing advanced data assimilation and machine learning methods for wildfire smoke transport forecasting. My academic training spans rigorous undergraduate preparation, elite graduate education at the African Institute for Mathematical Sciences, and advanced doctoral research funded by the National Science Foundation. My peer-reviewed publications, award-winning conference presentations, software contributions to Smoke3D, and leadership roles within both research and educational settings provide a clear record of success in areas directly tied to the proposed endeavor.

Federal agencies, including the National Science Foundation, the U.S. Naval Research Laboratory, and NASA researchers, have expressed strong interest in my work and its applicability to national forecasting and environmental resilience efforts. Expert evaluations from senior scientists and professors confirm that I possess rare and highly sought-after expertise in variational data assimilation, inverse problems, atmospheric transport modeling, and scientific machine learning, skills that are in short supply and of strategic importance to the United States.

Taken together, this evidence satisfies the second prong of the Dhanasar framework: I have the specialized education, research accomplishments, technical skills, and professional recognition required to make substantial contributions to the advancement of wildfire smoke forecasting systems in the United States.

Chapter Four

Waiving the Job Offer and Labor Certification Requirement Benefits the United States

The third prong of the NIW framework, as established in *Matter of Dhanasar*, requires a showing that, on balance, it would be beneficial to the United States to waive the job offer and labor certification requirements. In *Dhanasar*, USCIS concluded that the petitioner "would benefit the United States even assuming that other qualified U.S. workers are available" because of his proven record of contributing to work of national importance.

The evidence discussed in the previous chapters demonstrates that I satisfy this prong. The submitted evidence shows I have considerable experience and expertise in developing advanced data assimilation methods and model error estimation techniques for wildfire smoke transport forecasting. Work in this field is beneficial for the United States because it directly impacts the health of millions of Americans affected by wildfire smoke and supports federal agency missions in environmental protection, public health, and emergency management.

4.1 My Work Addresses an Urgent National Need

As demonstrated in Chapter 2, wildfire smoke forecasting is a matter of substantial merit and national importance. My work directly addresses federally documented gaps in operational forecasting capabilities through advanced model error estimation and data assimilation techniques. The National Science Foundation's investment of nearly $550,000 in the project I contribute to reflects federal recognition that these mathematical and computational advances are essential for improving national environmental prediction systems.

Dr. Mojtaba Sadegh emphasizes: ***"Ms. Sandra R. Babyale is an exceptional researcher whose work on wildfire smoke transport, machine learning, and model error estimation provides significant value to the United States. Her research advances mathematical and computational tools that protect public health and support national wildfire response and climate resilience.***

Her continued contributions will benefit federal agencies, research institutions, and the broader public." (See Exhibit A4)

4.2 Critical Workforce Shortage

Federal agencies have documented critical workforce shortages in the exact fields where my expertise lies. The U.S. Government Accountability Office specifically identified barriers to recruiting machine learning expertise for natural hazard modeling: ***"Workforce and resource needs create barriers to uptake of machine learning … the general federal pay scale limitations and workforce location requirements makes it difficult to compete with other sectors to hire and retain highly qualified staff with machine learning experience." (see Exhibit C3, pages 34-35)***

The Wildland Fire Mitigation and Management Commission Report to Congress confirms the urgency of addressing this workforce gap: ***"Federal investment is urgently needed to create a cross-trained year-round workforce that is focused on and tailored to mitigation, planning, and post-fire response and recovery" (see Exhibit C2, Supporting and Expanding the Workforce section)***

This federally documented shortage is corroborated by leading experts in my field. Professor Donna Calhoun adds: "***The US has long had to seek expertise in computational mathematics from abroad, exactly because these positions are difficult to fill with US citizens." (See Exhibit A3)***

Dr. Mojtaba Sadegh confirms: ***"There is a significant shortage of researchers with expertise in wildfire smoke transport modeling, advanced data assimilation, model error estimation, and scientific machine learning for environmental systems. Ms. Babyale's skill set directly fills this national workforce gap and supports ongoing federal efforts to strengthen environmental prediction and climate resilience." (See Exhibit A4)***

Dr. Hans Ngodock of the U.S. Naval Research Laboratory confirms the depth of this shortage: ***"It is very difficult to train or to find scientists in variational data assimilation. One of the***

reasons being that many pioneers in this field have retired, and most departments in the academia prefer teaching and using simpler methods for data assimilation." (See Exhibit A1)

These expert assessments, combined with federal agency findings, establish that my specialized combination of skills; variational data assimilation, model error estimation, inverse problems, and scientific machine learning, addresses a documented national workforce shortage that cannot be easily filled through traditional labor certification processes.

4.3 Contributions to U.S. Scientific and National Security

My research supports federal missions in environmental protection, public health, climate resilience, and emergency management. Professor Donna Calhoun states: ***"Sandra is at the beginning of her career and, with support, will be an asset to US scientific and national security for decades to come." (See Exhibit A3)***

Most significantly, Dr. Hans Ngodock of the U.S. Naval Research Laboratory explicitly expresses his intention to recruit me: ***"I will not hesitate to recruit Sandra upon her graduation. She will make significant contributions to our modeling and assimilation research efforts in support of the US Navy." (See Exhibit A1)***

This explicit recruitment interest from a senior federal researcher demonstrates that my expertise is not only rare but actively sought by federal agencies tasked with national security and environmental prediction missions. This federal agency interest provides powerful evidence that the traditional labor certification process, designed to protect U.S. workers, is unnecessary and impractical in my case, as federal employers are already seeking to recruit me specifically.

4.4 Strengthening the U.S. STEM Pipeline

My teaching and mentorship activities contribute to developing the next generation of American scientists and engineers. Professor Joe Champion states: ***"The United States' economic and national security is inextricably linked to leadership in science and technology, which in turn depends on a robust and growing STEM workforce. This workforce needs to be cultivated by mentors like Ms. Babyale." (See Exhibit A5)***

4.5 Unique Contributions That Cannot Be Replicated Through Labor Certification

The labor certification process is not well-suited to my situation for several reasons:

1. **Federally Documented Workforce Shortage:** Both the GAO and the Wildland Fire Commission have documented critical shortages of researchers with machine learning expertise for environmental modeling. The GAO specifically notes that ***"federal pay scale limitations and workforce location requirements makes it difficult to compete with other sectors to hire and retain highly qualified staff" (see Exhibit C3, pages 34-35).*** This demonstrates that the traditional labor market test is impractical, federal agencies cannot easily recruit qualified U.S. workers for these specialized positions even when they actively seek them.

2. **Specialized Expertise:** My combination of expertise in variational data assimilation, model error estimation, and machine learning for atmospheric applications is rare and cannot easily be found through traditional recruitment. Chapter 3 documents that multiple experts have confirmed this shortage exists and that pioneers in variational data assimilation have retired without sufficient replacement in the U.S. academic pipeline.

3. **Ongoing Research Continuity:** My NSF-funded research is at a critical stage, with a peer-reviewed publication already advancing the field and additional manuscripts in preparation. Disruption of this federally funded project would harm the national investment already made and delay progress on essential forecasting improvements.

4. **Federal Agency Interest:** Dr. Ngodock, a senior scientist at the U.S. Naval Research Laboratory with documented authority to recruit PhD graduates, has committed without reservation to recruiting me upon graduation, stating: ***"I will not hesitate to recruit Sandra upon her graduation. She will make significant contributions to our modeling and assimilation research efforts in support of the US Navy." (See Exhibit A1).*** This commitment from a federal defense research institution demonstrates that federal employers have identified a specific operational need for my expertise and are actively seeking to add me to the federal workforce to fulfill national security missions.

4.6 Balance of Interests

Considering:

- The urgent national importance of wildfire smoke forecasting, as demonstrated by the substantial merit and national importance analysis in Chapter 2
- The federally documented workforce shortage in machine learning and data assimilation expertise *(See Exhibit C3, pages 34-35; Exhibit C2).*
- My peer-reviewed publication in a leading journal (SIAM/ASA JUQ) *(See Exhibit D1)*
- My seven peer-reviewed conference presentations, including one award-winning poster *(See Exhibits D4 - D16)*
- My NSF-funded research contributions ($550,000 federal investment) *(See Exhibit D3)*
- My software contributions (Smoke3D)
- The explicit interest from the U.S. Naval Research Laboratory in recruiting me
- My contributions to STEM education and the American workforce
- My position as a member of the professions holding an advanced degree with exceptional qualifications documented in Chapter 3

On balance, it would benefit the United States to waive the requirements of a job offer and labor certification. My contributions are of such value that they would benefit the United States even assuming other qualified U.S. workers are available.

The labor certification process would be impractical and contrary to national interests in this case because:

- Federal agencies have documented they struggle to recruit and retain workers with my specialized skills even when actively seeking them
- My ongoing NSF-funded research requires continuity to fulfill the federal investment already made
- Federal researchers have explicitly expressed intent to recruit me, demonstrating that the labor market test is unnecessary

- My dual expertise in both classical data assimilation and machine learning fills a documented gap that federal agencies have identified as critical for advancing environmental forecasting systems

4.7 Expert Recommendations

All expert letter writers recommend approval of my NIW petition:
Dr. Hans Ngodock of the U.S. Naval Research Laboratory concludes: ***"I therefore highly recommend the approval of her immigration application for a EB-2 NIW." (See Exhibit A1)***

Professor. Jodi Mead states: ***"I strongly recommend that the U.S. Citizenship and Immigration Services approve Sandra Babyale's National Interest Waiver." (See Exhibit A2)***

Professor Donna Calhoun affirms: ***"I am happy to give her my highest recommendation for permanent residency in the US." (See Exhibit A3)***

Dr. Mojtaba Sadegh concludes: ***"I strongly recommend approval of her National Interest Waiver petition." (See Exhibit A4)***

Professor. Joe Champion states: ***"I recommend approval of her National Interest Waiver petition with the utmost enthusiasm and without reservation." (See Exhibit A5)***

Conclusion

The evidence provided demonstrates that:

1. **My proposed endeavor holds substantial merit and national importance.**
2. **I am exceptionally well-positioned to advance this endeavor**, supported by my academic training, NSF-funded research, peer-reviewed publications, technical software contributions, professional leadership, and strong expert testimony.
3. **Waiving the job offer and labor certification requirement will benefit the United States**, given the national urgency of wildfire smoke forecasting, the scarcity of researchers with my expertise, federal agency interest in my work, and my contributions to the STEM pipeline.

Accordingly, I respectfully request approval of my petition under the Employment-Based Second Preference, National Interest Waiver (EB2-NIW) category.

Respectfully Submitted,

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